MURATOV ARTEM

Compiler Development & Low-Level Optimization

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EXPERIENCE

Summer Internship at Manpower (a Huawei project)

Position: Assistant Engineer (Binary Optimizer Team)

01 Jun 2025 – 22 Aug 2025

- Developed an optimization pass for an ARM (AArch64) binary optimizer using an LLVM-based internal framework.
- The pass replaces simple comparison patterns with `CSEL` (Conditional Select) instructions, reducing branch mispredictions and improving code density.
- Collaborated with senior engineers during code reviews to integrate the pass into the main optimizer pipeline.

EDUCATION

Introduction to industrial programming in C++ course, 2025 – present

Current average: 8.0/10

MIPT course, Lecturer: Konstantin Vladimirov

Introduction to tensor compilers course, 2026 – present

MIPT and Sber course, Lecturers: Alexey Eganyan, Vladislav Loznovenko

System programming and compiler technology, course 2024 – 2025

Final grade: 8.0/10

MIPT course, Lecturer: Ilya Dedinsky

Bachelor of Radio Engineering and Computer Technology, 2024 – present

GPA: 7.27/10 (current, as of Mar 2026)

Moscow Institute of Physics and Technology

MAIN PROJECTS

Tensor Compiler (C++, MLIR, LLVM), 2026

- Developing a compiler for neural network inference, focusing on graph optimization and code generation.
- Current Focus: Implementing a lowering pipeline from a MLIR Dialect to LLVM IR, and finally to native Assembly.
- Designed a computational graph backend capable of parsing ONNX models (via Protobuf) and supporting 6+ NN operators (Conv, Gemm, MatMul, etc.).

ParaCL Compiler (C++, LLVM, Flex/Bison), 2025–2026

- Designed a multi-pass compiler architecture: LALR(1) parser (Bison), typed AST with Visitor pattern, and an LLVM IR backend utilizing `llvm::IRBuilder<>` for native binary generation.
- Implemented robust scope management and a detailed error recovery system with precise source location tracking and syntax correction suggestions.
- Developed a dual-mode execution engine supporting both AST interpretation and optimized native compilation via LLVM's `opt` pipeline.
- Ensured code reliability with Google Test, writing unit and end-to-end tests for complex control flow, bitwise operations, and chained assignments.

Implementation of bitonic sort algorithm with GPU (C++, OpenCL), 2026

- Developed an OpenCL-based GPU sorting algorithm achieving up to 8x speedup on 1M elements over std::sort on CPU (tested on Intel(R) UHD Graphics).
- Optimized kernel by using local memory and workgroup-level parallelism.

High-performance word search engine with Hash Table (C/C++, x86-64 ASM, Intrinsics), 2025

- Developed a hash table search engine processing 75M queries on 20K unique words, achieving 96% speedup (14.36s $\rightarrow$ 7.33s) against -O3 baseline.
- Optimized hot paths using SIMD intrinsics (__m128i) for string comparison and hardware CRC32 instructions for hashing, guided by Perf profiling.

Compiler for a custom programming language (C/C++, x86-64 ASM), 2025

- Developed a multi-pass compiler: recursive-descent frontend, IR with optimizations (constant folding, neutral expression removal), and backend targeting x86-64 and a simulated custom CPU.
- Generated assembly code that passes all provided test cases.

TECHNICAL SKILLS

Programming Languages

C++ (modern), C, LLVM IR, MLIR, Assembly (x86-64, ARM basics), Python, OpenCL, GLSL

Core Competencies

Compiler Design (LLVM, MLIR), Computer Architecture, Performance Profiling (Perf), Low-level Optimization, SIMD.

Languages

Russian (Native), English (B2)

Interests

Member of the MIPT table tennis team and football player. Reading classic literature and self-development books.